MACHINE LEARNING BASED EARLY PREDICTION OF FISH DISEASES THROUGH WATER QUALITY ANALYSIS

ABSTRACT

Aquaculture plays a significant role in food production and the economy, but fish diseases continue to be a major challenge for fish farmers. One of the primary causes of these diseases is poor water quality, which directly affects the health and growth of fish. This project presents a machine learning-based approach for the early detection of fish health risks using important water quality parameters such as pH, dissolved oxygen (DO), temperature, turbidity, conductivity, nitrate, biological oxygen demand (BOD), and bacterial indicators. The proposed solution offers an efficient, cost-effective, and intelligent decision support system for aquaculture management. By enabling early detection of potential health risks, it helps improve fish survival, reduce disease outbreaks, enhance water quality management, and support sustainable fish farming practices. In addition to prediction, the proposed system identifies the water quality parameters responsible for the disease risk and provides recommendations and preventive measures to help farmers maintain a healthy aquatic environment

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